MEC6602E : Homework 2,
25% of final grade
Due date as per course Syllabus

Maximum 5 pages and can use appendices if you want to provide details that are secondary but whose understanding is important to your analysis. You will be judged on both content (75%) and form (25%). Content refers to the exercises requested, the precision of the analytical and numerical calculations (25 points), and the quality of the analysis of your results. **Implement the algorithm presented in class. Any deviation from these algorithms should be justified.** Form refers to style, clarity of presentation and quality of figures/tables. The detailed evaluation grid is:

Evaluation table

	points
exercises	5
precision	5
analysis	5
form	5

2D steady Euler equations

The 2D Euler equations are available in standard textbooks or in scientific literature as provided on Moodle. Write a CFD code that solves the 2D Euler solutions around an airfoil (NACA0012) for **subsonic**, **transonic** and **supersonic** conditions. The mesh topology around the airfoil is a single-block O-grid cut at the trailing edge, to ease the topological complexity of the code. The meshes are provided, available on Moodle. The code takes as input a mesh, a Mach number and an angle-of-attack. As output, it writes the CL, CD and CM (around the quarter-chord point) on the airfoil boundary, a field solution file and a file for visualizing the pressure on the surface of the airfoil. Points will be awarded if the code is able to run on the largest grid provided. **Please report on the convergence and execution time of the software**.

Part 1: Implement a finite volume scheme with a second order spatial accuracy (central scheme with artificial dissipation or Roe scheme) and an explicit Runge-Kutta scheme.
Part 2: Add anything you want to your code. At the minimum, add one thing: Residual smoothing, implicit scheme, multigrid, etc.
Part 3: Perform a numerical analysis on the behavior and order of convergence of the solver. The baseline comparison is Vassberg and Jameson 2010 paper. Points will be attributed on the basis of the largest grid the software was able to handle (up to 2049 X 2049). Select one of the flow condition in Vassberg and Jameson 2010 to perform this analysis.
Part 4: Analyse soution in subsonic, transonic and supersonic condition on a sufficiently refine mesh
Part 5: Implement the features required to execute simulation at fix lift coefficient. Compute the flow at Mach 0.8 and CL 0.4 on a mesh sufficiently refined.

To simplify the development use the codes provided on Moodle (in C language) to:

- Read the mesh file;
- Output field solution in tecplot ASCII format;
- Facets loop example;
- Cell center loop example.

  If you want to make use of a software developed in another course, you are required to provide a version of the initial code no later than september 29th. In such case an agreement can be made on what you will implement.

Reference:

1) Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, *47*(4), 1152-1166.

2) Blazek, J. (2015). *Computational fluid dynamics: principles and applications*. Butterworth-Heinemann.

3) Witherden, F. D., Jameson, A., & Zingg, D. W. (2017). The design of steady state schemes for computational aerodynamics. In *Handbook of Numerical Analysis* (Vol. 18, pp. 303-349). Elsevier.